



Lesson-by-Lesson Guide

From Zero to Pro - The Self-Taught Chef Program

12 modules - 48 lessons - the complete teaching companion

Read alongside the videos, then cook.

How to use this guide

This guide is the written companion to the program. Each of the twelve modules is broken into four lessons, and each lesson follows the same rhythm so you always know where you are: a one-line **objective**, a short **teaching** section explaining the what and why, a list of **key points** to remember, a **step-by-step** technique where one applies, the **common mistakes** that trip people up, and a **practice** task to cement the skill in your own kitchen.

The best way to use it

- Read the module introduction, then watch the linked videos for that module.
- Read each lesson before you cook it; keep the guide open while you work.
- Do the practice task at the end of every lesson - reading is not cooking.
- Log dishes on the interactive Recipe Card and check skills off the Progress Tracker.
- Move to the next module only when the current lesson's skills feel automatic.

Pair this guide with the module slide decks (in the Slide-Decks folder) for a lecture-style overview, and the Reference Library for quick lookups of cuts, temperatures, and ratios.

MODULE 01 - Weeks 1-2

Kitchen Foundations: Safety, Sanitation & Setup

Big idea: A safe, clean, organized kitchen is the platform every other skill stands on.

Before you cook anything, you set up a kitchen that is safe, clean, and organized like a professional line.

What you'll be able to do

- Practice the core rules of food safety: the temperature danger zone, cross-contamination, and FIFO storage.
- Set up an efficient, sanitary workstation and clean as you go.
- Identify the essential tools and how to maintain them.
- Read a recipe like a pro: scaling, mise en place, and timing.

Watch alongside this module (free)

[Internet Shaquille \(efficiency & fundamentals\)](#)

[Adam Ragusea \(practical home technique\)](#)

[USDA FSIS - Safe Minimum Internal Temperature Chart](#)

[FoodSafety.gov - Safe Minimum Internal Temperatures](#)

Lesson 1.1 - Food Safety and the Temperature Danger Zone

Objective: Keep food out of the temperature range where bacteria multiply fastest.

Most food-borne illness comes from time and temperature, not from dirt. Bacteria multiply fastest between 40 and 140 degrees F (4 to 60 deg C) - the danger zone. Your job is to move food through that range quickly and not let it linger there.

The working rule is simple: keep cold food cold (below 40 deg F), keep hot food hot (above 140 deg F), and never leave perishable food at room temperature for more than two hours total. Cook to safe internal temperatures, cool leftovers fast, and reheat to 165 deg F.

Key points

- Danger zone: 40-140 deg F (4-60 deg C); limit total time there to under 2 hours.
- An instant-read thermometer is the only reliable doneness test - buy one.
- Thaw in the fridge, in cold water, or in the microwave - never on the counter.
- Cool hot leftovers quickly (shallow containers) before refrigerating.

Technique, step by step

1	Chill	Keep your fridge at or below 40 deg F; verify with a fridge thermometer.
2	Separate	Store raw meat on the bottom shelf so it cannot drip onto other food.
3	Cook	Use a thermometer to hit safe internal temps (see Reference Library).
4	Cool	Get leftovers below 40 deg F within a couple hours; divide into shallow pans.

Common mistakes to avoid

- Trusting color or time instead of a thermometer.
- Thawing meat on the counter all afternoon.
- Cramming a huge hot pot straight into the fridge, where it cools too slowly.

PRACTICE

Put a thermometer in your fridge and check it. Cook one protein this week strictly by internal temperature, not by guessing.

Lesson 1.2 - Sanitation and Cross-Contamination

Objective: Stop harmful bacteria from moving between raw foods, surfaces, and ready-to-eat foods.

Cross-contamination is the transfer of bacteria from one food (usually raw meat, poultry, seafood, or eggs) to another food, a surface, or your hands. It is the second great cause of kitchen illness, and it is entirely preventable with a system.

The system is separation and cleaning. Use distinct cutting boards for raw protein and for produce, wash hands and tools after handling raw items, and sanitize surfaces. Handwashing - twenty seconds with soap - is the single most effective thing you can do.

Key points

- Use separate boards: one for raw protein, one for produce and ready-to-eat foods.
- Wash hands 20 seconds with soap after handling raw meat, eggs, or trash.
- Clean removes visible soil; sanitizing kills microbes - do both on surfaces.
- Be aware of the top allergens when cooking for others (e.g., nuts, shellfish, dairy).

Common mistakes to avoid

- Slicing salad on the same board you just cut raw chicken on.
- Wiping hands on a towel instead of washing them.
- Tasting with a spoon and double-dipping it back into the pot.

PRACTICE

Designate and label two cutting boards today. Cook a meal with raw chicken and a fresh salad without ever letting them share a surface.

Lesson 1.3 - Setting Up Your Station and the Mise en Place Mindset

Objective: Organize a workspace so cooking is calm execution, not a scramble.

Professionals do not improvise their setup; they standardize it. Mise en place - 'everything in its place' - means all ingredients measured and prepped, all tools within reach, a clear cutting area, a trash bowl, and a stack of towels before any heat goes on.

Clean as you go. A clear station is a safe, fast station. The habit of wiping down, consolidating scraps, and washing idle tools while something simmers is what separates a relaxed cook from a frantic one.

Key points

- Prep everything before you turn on a burner; reading mid-cook costs you control.
- Keep a 'garbage bowl' on the counter to keep your board clear.
- Set a damp towel under your cutting board so it cannot slip.
- Clean as you go - idle minutes are for washing, wiping, and resetting.

Common mistakes to avoid

- Starting to cook with half the ingredients still in the fridge.
- Letting scraps and dirty tools pile up until the station is unusable.
- Hunting for a tool while food burns.

PRACTICE

Cook one dinner where you do 100 percent of the prep first, arrange your station, and only then start cooking. Notice how much calmer it feels.

Lesson 1.4 - Reading, Scaling, and Timing a Recipe

Objective: Read a recipe like a plan, not a story, and run it on a timeline.

Before cooking, read the entire recipe twice. The first pass tells you what you are making; the second pass reveals the timeline - what has to happen in parallel, what can be done ahead, and where the pressure points are. Pull out the verbs and the temperatures.

Scaling is arithmetic, but it has limits: seasoning and cook times do not always scale linearly, and pan size matters. Weighing ingredients in grams makes scaling reliable, especially in baking. Build a backward timeline from when you want to eat.

Key points

- Read the whole recipe twice before starting; note the prep hidden inside the steps.
- Work by weight (grams) when you can - it scales cleanly and travels between recipes.
- Build a backward timeline from serving time so components finish together.
- Salt and cook times do not scale perfectly; taste and check, do not just multiply.

Common mistakes to avoid

- Discovering 'marinate overnight' an hour before dinner.
- Doubling a recipe in a pan too small to brown - everything steams.
- Following volumes for baking instead of weighing.

PRACTICE

Take a favorite recipe, rewrite it as a backward timeline, and cook from your timeline instead of the original.

MODULE 02 - Weeks 3-4

Knife Skills & Mise en Place

Big idea: Knife skills are the highest-leverage skill in cooking; build safe, consistent cuts before speed.

The single highest-leverage skill in cooking. Speed comes later; first you build safe, consistent cuts.

What you'll be able to do

- Hold a chef's knife with the pinch grip and use the claw guiding hand.
- Execute the standard cuts: slice, julienne, brunoise, dice, chiffonade, mince.
- Break down an onion, carrot, and herbs cleanly and with minimal waste.
- Hone and understand sharpening; care for your blade.

Watch alongside this module (free)

[Gordon Ramsay - How to Master 5 Basic Cooking Skills](#)

[Jacques Pepin - Techniques & More \(KQED\)](#)

[J. Kenji Lopez-Alt \(food science & technique\)](#)

Lesson 2.1 - The Knife and the Grip

Objective: Hold the knife with control using the pinch grip and the guiding claw.

An 8 to 10 inch chef's knife handles roughly ninety percent of kitchen tasks. Control starts with the grip: pinch the blade - not the handle - between your thumb and the side of your bent index finger, just ahead of the bolster. The knife becomes an extension of your hand.

Your other hand is the claw: fingertips curled under, knuckles forward, so the flat of the blade rides against your knuckles and your fingertips can never reach the edge. A sharp knife is safer than a dull one because it goes where you aim it.

Key points

- Pinch grip on the blade gives control; gripping the handle alone is weak.
- Claw hand: knuckles guide the blade, fingertips tuck safely away.
- Let the knife's weight and curve do the work; do not force it.
- A sharp knife is a safe knife - it does not slip.

Technique, step by step

1	Pinch	Thumb on one side of the blade, curled index finger on the other.
2	Claw	Curl guiding fingertips under; rest the blade's flat on your knuckles.
3	Rock	Keep the tip on the board and rock the blade through the cut.
4	Walk	Slide the claw hand back to set the next slice width.

Common mistakes to avoid

- Choking up on the handle with no blade pinch.

- Pointing fingertips at the blade instead of tucking them.
- Sawing back and forth instead of rocking smoothly.

PRACTICE

Spend ten minutes just holding the pinch grip and rocking the empty knife, then slice a few scallions using the claw.

Lesson 2.2 - The Standard Cuts

Objective: Produce the classic cuts at consistent dimensions.

Recipes assume a shared vocabulary of cuts because size controls how fast and evenly food cooks. The core cuts run from large to small: slice, rondelle, batonnet (1/4 inch sticks), julienne (1/8 inch matchsticks), then the dice that come from them - small dice from batonnet, brunoise from julienne.

Two more finish the set: chiffonade (thin ribbons of leafy herbs or greens, made by stacking, rolling, and slicing) and mince (very fine, irregular, for garlic and shallots). Consistency matters more than speed - even pieces cook evenly.

Key points

- Julienne = 1/8 inch matchsticks; brunoise = 1/8 inch dice from julienne.
- Batonnet = 1/4 inch sticks; small dice = 1/4 inch cube from batonnet.
- Square off rounded vegetables first so cuts are even and stable.
- Even size beats speed - uniform pieces cook at the same rate.

Technique, step by step

1	Square off	Trim a round vegetable into a rectangular block for flat, stable sides.
2	Plank	Slice the block into even slabs.
3	Stick	Stack slabs and cut into sticks (julienne or batonnet).
4	Dice	Line up the sticks and cut across to make cubes.

Common mistakes to avoid

- Cutting on rounded surfaces that roll and slip.
- Mixing wildly different sizes so some pieces burn and others stay raw.
- Pressing leafy herbs into a paste instead of slicing ribbons cleanly.

PRACTICE

Turn one carrot into clean batonnet, then dice half of it. Chiffonade a few basil leaves. Judge yourself on evenness.

Lesson 2.3 - Breaking Down Aromatics

Objective: Cut onion, garlic, shallot, and herbs cleanly with minimal waste.

Most savory cooking starts with aromatics, so dicing an onion is a foundational move. Halve it pole to pole through the root, peel but keep the root intact as a handle, then make vertical cuts toward the root and cross-cut to release an even dice. The root holds the layers together.

Garlic can be smashed under the flat of the blade to loosen its skin, then minced. Shallots are diced like tiny onions. Herbs are either picked and chopped (parsley, cilantro) or chiffonaded (basil, mint). Pile, do not scatter, so your board stays workable.

Key points

- Cut onions pole to pole and keep the root as a handle to hold layers together.
- Smash garlic with the flat of the blade to skin it, then mince.
- Dice shallots like miniature onions for fine, even aromatics.
- Gather scraps into your garbage bowl to keep the board clear.

Technique, step by step

1	Halve	Cut the onion pole to pole through the root and peel.
2	Vertical	Make evenly spaced cuts toward (not through) the root.
3	Cross-cut	Slice across to drop an even dice; spacing sets the size.
4	Finish	Discard the root and gather the dice.

Common mistakes to avoid

- Cutting through the root so the onion falls apart mid-dice.
- Mincing garlic so fine and early it scorches in the pan.
- Chopping wet herbs that bruise into a dark paste.

PRACTICE

The five-onion drill: dice five onions aiming for even pieces, and save them for soups and stocks. Nothing wasted.

Lesson 2.4 - Honing, Sharpening, and Knife Care

Objective: Keep an edge that is sharp, aligned, and safe.

A blade dulls in two ways: the fine edge rolls out of alignment (fixed by honing) and the metal actually wears down (fixed by sharpening). Hone with a steel before most sessions - a few light strokes per side realigns the edge in seconds and removes no metal.

Sharpen on a whetstone every few weeks to months, depending on use, to grind a fresh edge. Care matters too: hand-wash and dry knives immediately, never put them in the dishwasher, store them on a magnetic strip or in a guard, and cut on wood or plastic, never glass or stone.

Key points

- Honing realigns the edge (do it often); sharpening removes metal (do it seldom).
- Hand-wash and dry knives right away; the dishwasher ruins edges and handles.
- Cut only on wood or plastic - glass and stone destroy the edge.
- Store on a magnet strip or in-drawer guard, never loose in a drawer.

Common mistakes to avoid

- Never honing, then wondering why the knife slips and crushes.
- Tossing a good knife in the dishwasher.
- Trying to catch a falling knife - step back and let it drop.

PRACTICE

Hone your knife before your next three cooking sessions and notice the difference. If you have a whetstone, sharpen once and compare.

MODULE 03 - Weeks 5-6

Heat & Cooking Methods

Big idea: Every dish is an application of heat; learn the methods and you can cook almost anything without a recipe.

Every dish is an application of heat. Learn the methods and you can cook almost anything without a recipe.

What you'll be able to do

- Distinguish dry-heat (saute, sear, roast, grill, fry) from moist-heat (steam, poach, simmer, braise) methods.
- Control a pan: preheating, fond, deglazing, and the Maillard reaction.
- Choose the right method for a given ingredient.
- Use a thermometer and your senses to judge doneness.

Watch alongside this module (free)

[Ethan Chlebowski \(the 'why' behind cooking\)](#)

[J. Kenji Lopez-Alt \(food science & technique\)](#)

[America's Test Kitchen \(tested technique\)](#)

Lesson 3.1 - How Heat Moves and Why Food Browns

Objective: Understand conduction, convection, radiation, and the browning reactions.

Heat reaches food three ways: conduction (direct contact, like a steak on a pan), convection (a moving fluid, like a hot oven or boiling water), and radiation (energy waves, like a broiler or grill). Knowing which is at work tells you how to control the cook.

Browning is flavor. The Maillard reaction - between proteins and sugars above roughly 300 deg F - creates the deep savory crust on seared meat and toasted bread. Caramelization is the browning of sugars alone. Both need a dry, hot surface; water capped at 212 deg F cannot brown.

Key points

- Conduction = contact, convection = moving fluid, radiation = waves.
- Maillard browning (proteins + sugars) builds savory crust above ~300 deg F.
- Caramelization is sugars browning alone - sweet and nutty.
- Wet food steams and will not brown until the surface dries.

Common mistakes to avoid

- Crowding a pan so food steams in its own moisture and never browns.
- Adding salt to meat right before searing, drawing moisture to the surface.
- Expecting color from a method that stays at or below boiling.

PRACTICE

Pat a piece of chicken or tofu very dry, sear half in a hot dry-ish pan and half in a crowded cool pan. Compare the crust.

Lesson 3.2 - Dry-Heat Methods

Objective: Control sauteing, searing, roasting, grilling, and frying.

Dry-heat methods brown food and build flavor. Sauteing cooks small or thin pieces fast in a little fat over high heat, keeping the food moving. Searing develops a crust on larger cuts - preheat the pan, add a thin film of oil, and let the food sit undisturbed until it releases.

Roasting surrounds food with hot, dry oven air; high heat browns exteriors while interiors cook through. Grilling and broiling use radiant heat for char. Frying cooks in hot fat (325 to 375 deg F), which is fast and, when the oil is hot enough, surprisingly non-greasy.

Key points

- Preheat the pan before adding oil and food; a cold pan means no sear.
- Let seared food sit undisturbed - it releases from the pan when crusted.
- Do not crowd; cook in batches so heat stays high and food browns.
- Fry at 325-375 deg F; too-cool oil makes greasy food.

Technique, step by step

1	Preheat	Heat the empty pan, then add a thin film of oil until it shimmers.
2	Lay away	Place food down away from you; do not move it.
3	Wait	Let the crust form; the food will release when ready.
4	Flip once	Turn a single time for an even, deep sear.

Common mistakes to avoid

- Flipping constantly so no crust ever forms.
- Overcrowding the pan and steaming the food.
- Frying in oil that is too cold, soaking the food in grease.

PRACTICE

Sear a protein: preheat, oil, lay it away, and do not touch it until it releases. Build a crust you can hear and see.

Lesson 3.3 - Moist-Heat and Combination Methods

Objective: Steam, poach, simmer, and braise with control.

Moist-heat methods are gentle and do not brown. Steaming cooks food in vapor and preserves texture and nutrients. Poaching holds food in liquid at 160 to 180 deg F - barely a shiver - ideal for eggs, fish, and fruit. Simmering (185 to 205 deg F) and boiling (212 deg F) cook pasta, grains, and stocks.

Combination methods get the best of both: braising and stewing sear food first for flavor, then cook it slowly, partly submerged in liquid. Low, slow heat melts the tough collagen in cheap cuts into silky gelatin - which is why a braise turns chuck or short rib meltingly tender.

Key points

- Poach at 160-180 deg F (a bare shiver), not a rolling boil.
- Simmer = gentle bubbles (~185-205 deg F); boil = rolling (212 deg F).
- Braise = sear, then slow-cook partly submerged in liquid.
- Low and slow turns tough collagen into tender gelatin.

Technique, step by step

1	Sear	Brown the meat well on all sides for flavor and color.
2	Build	Soften aromatics, deglaze, add just enough liquid to come partway up.
3	Cover	Lid on, low oven or low burner, until fork-tender.
4	Reduce	Strain and reduce the liquid into a glossy sauce.

Common mistakes to avoid

- Boiling a braise hard, which makes meat dry and stringy.
- Drowning a braise in liquid so it boils instead of braises.
- Poaching delicate fish in water that is actually boiling.

PRACTICE

Braise a tough cut: sear it, add aromatics and a little liquid, cover, and cook low until a fork twists easily. Reduce the liquid to sauce.

Lesson 3.4 - Doneness, Resting, and the Pan Sauce

Objective: Judge doneness, use carryover and resting, and build a sauce from fond.

Doneness is about internal temperature, and food keeps cooking after it leaves the heat - carryover can raise a roast 5 to 10 degrees. So pull food a few degrees early and let it rest; resting lets juices redistribute so they stay in the meat instead of flooding the board.

While the protein rests, build a pan sauce from the fond - the browned bits stuck to the pan. Pour off excess fat, add aromatics, deglaze with wine or stock to dissolve the fond, reduce, and finish with a knob of butter. That is restaurant flavor from what looked like a dirty pan.

Key points

- Use a thermometer; pull a few degrees early for carryover.
- Rest meat so juices redistribute - slicing too soon loses moisture.
- Fond (browned bits) is the base of a great pan sauce.
- Deglaze, reduce, then finish with butter for body and shine.

Technique, step by step

1	Rest	Move the protein to a board; tent loosely while you make the sauce.
2	Pour off	Drain excess fat, leaving the fond and a little fat behind.

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|---|--------------------------|--|
| 3 | Deglaze | Add liquid and scrape up the fond as it dissolves. |
| 4 | Reduce and finish | Simmer to concentrate, then swirl in cold butter off heat. |

Common mistakes to avoid

- Cutting into meat immediately, spilling the juices.
- Scrubbing away the fond before you can make a sauce from it.
- Boiling butter into the sauce so it splits instead of emulsifying.

PRACTICE

Sear a protein to a target temp, rest it, and build a pan sauce from the fond while it rests. Taste before and after the butter.

MODULE 04 - Weeks 7-8

Stocks, Soups & the Five Mother Sauces

Big idea: Stocks, roux, and the five mother sauces are the classical backbone of most Western cooking. The classical French backbone that underlies most Western cooking. Master ratios and you master sauce.

What you'll be able to do

- Make clear stock and understand the roux.
- Produce the five mother sauces and several derivatives.
- Build soups: pureed, brothy, and creamed.
- Balance, season, and adjust a sauce to finish.

Watch alongside this module (free)

[Jacques Pepin - Techniques & More \(KQED\)](#)

[The Complete Pepin Archive \(Jacques Pepin Foundation\)](#)

[Brian Lagerstrom \(ex-restaurant pro, approachable\)](#)

Lesson 4.1 - Stocks: The Foundation of Flavor

Objective: Make a clear, flavorful stock from bones and aromatics.

Stock is the liquid foundation under soups, sauces, and braises. White stock starts with raw bones; brown stock starts with roasted bones for deeper color and flavor. Either way: cover bones and aromatics (mirepoix) with cold water, bring up slowly, and simmer gently - never boil, which makes stock cloudy and greasy.

Skim the foam and fat that rise, keep the heat at a bare simmer, and strain. A good stock is built on patience and low heat. Reduce a finished stock far enough and it becomes a glace - a concentrated, syrupy flavor bomb you can freeze.

Key points

- Start in cold water and heat slowly for a clear stock.
- Simmer gently and skim; a hard boil makes it cloudy and greasy.
- Mirepoix (onion, carrot, celery, 2:1:1) is the standard aromatic base.
- Reduce hard to make glace - intense, freezable flavor.

Technique, step by step

1	Combine	Cover bones and mirepoix with cold water.
2	Heat slowly	Bring up to a bare simmer, never a boil.
3	Skim	Remove foam and fat as they rise to the top.
4	Strain	After several hours, strain and cool quickly.

Common mistakes to avoid

- Boiling the stock, which turns it cloudy and greasy.
- Salting heavily early - stock is a base and gets reduced later.
- Skipping the skim, leaving scummy, off flavors.

PRACTICE

Save a roast chicken carcass and make stock: cold water, mirepoix, gentle simmer, skim, strain. Reduce a cup of it to a syrupy glace.

Lesson 4.2 - Roux and Thickening

Objective: Cook roux to the right color and thicken liquids smoothly.

Roux is equal parts fat and flour by weight, cooked together to thicken sauces. Cook it briefly for white roux (bechamel), a bit longer for blond (veloute), and longer still for brown roux (espagnole), which is nuttier but thickens less because cooking breaks down its starch.

To thicken without lumps, combine roux and liquid at different temperatures - hot roux into cold liquid or cold roux into hot liquid - and whisk constantly as it comes up to a simmer. The sauce thickens as the starch swells. Other thickeners include cornstarch slurry and pure reduction.

Key points

- Roux = 1:1 fat to flour by weight.
- White, blond, and brown roux: darker = more flavor but less thickening power.
- Avoid lumps: contrast temperatures and whisk constantly.
- Cook out the raw-flour taste before judging the sauce.

Technique, step by step

1	Melt	Melt fat (usually butter) over medium heat.
2	Stir in flour	Add equal weight of flour; stir into a smooth paste.
3	Cook	Cook to the color you want, stirring.
4	Add liquid	Whisk in liquid gradually and bring to a simmer to thicken.

Common mistakes to avoid

- Dumping cold roux into cold liquid - it clumps.
- Not cooking out the flour, leaving a pasty raw taste.
- Expecting dark roux to thicken like white roux - it will not.

PRACTICE

Make a small white roux and whisk in milk to a smooth bechamel. Note exactly how much milk a given weight of roux thickens.

Lesson 4.3 - The Five Mother Sauces

Objective: Produce the five classical mother sauces and a derivative of each.

Escoffier organized French sauces into five mothers, each a base for many derivatives. Bechamel is milk thickened with white roux. Veloute is white stock with blond roux. Espagnole is brown stock with brown roux and tomato. Tomato sauce is built from tomatoes and aromatics. Hollandaise is an emulsion of egg yolk, butter, and acid.

Learn the mother and you unlock its children: bechamel plus cheese is mornay; hollandaise plus tarragon reduction is bearnaise; espagnole reduced with stock is demi-glace. The ratios and method carry across, so a handful of techniques produces dozens of sauces.

Key points

- Bechamel = milk + white roux.
- Veloute = white stock + blond roux; Espagnole = brown stock + brown roux + tomato.
- Tomato = tomatoes + aromatics; Hollandaise = yolk + butter + acid emulsion.
- Derivatives are small additions to a mother (e.g., mornay = bechamel + cheese).

Common mistakes to avoid

- Breaking hollandaise by adding butter too fast or letting it get too hot.
- Under-seasoning a bechamel - milk sauces need salt and nutmeg.
- Skipping the cook-out so the sauce tastes of raw flour.

PRACTICE

Make a bechamel and turn it into a mornay with grated cheese. Write down the ratio of roux to milk to cheese that you used.

Lesson 4.4 - Soups, Emulsions, and Rescues

Objective: Build soups, understand emulsions, and fix a broken sauce.

Most soups are one of three shapes: brothy (clear liquid with garnish), pureed (cooked vegetables blended smooth), or creamed (pureed and enriched with dairy). All start the same way - sweat aromatics, add liquid, simmer, then season and adjust at the end.

An emulsion suspends fat in water-based liquid (or vice versa); hollandaise and vinaigrette are examples. They break when the fat goes in too fast or the temperature is wrong. To rescue a broken emulsion, start fresh with a little warm water or a yolk and whisk the broken sauce into it slowly, rebuilding the suspension.

Key points

- Soup shapes: brothy, pureed, creamed - all begin by sweating aromatics.
- Season soups at the end and taste repeatedly as they reduce.
- Emulsions break from speed or wrong temperature.
- Rescue a break by re-emulsifying slowly into a fresh base.

Technique, step by step

1	Sweat	Soften aromatics in fat without browning.
2	Simmer	Add liquid and main ingredients; cook until tender.
3	Blend or leave	Puree for a smooth soup, or leave brothy.
4	Finish	Adjust salt and acid; enrich with cream if desired.

Common mistakes to avoid

- Adding cream and then boiling hard, which can curdle it.
- Pouring oil into a vinaigrette all at once so it will not hold.
- Forgetting acid - a squeeze of lemon often 'fixes' a flat soup.

PRACTICE

Make a pureed vegetable soup. Then deliberately break a small hollandaise and rescue it by whisking it into a fresh yolk.

MODULE 05 - Week 9

Eggs, Dairy & Breakfast Mastery

Big idea: Eggs are the classic test of a cook; gentle heat and timing are everything.

Eggs are the classic test of a cook. Gentle heat control and timing are everything.

What you'll be able to do

- Cook eggs every classic way: scrambled (French & American), omelet, poached, fried, boiled.
- Understand custards, curds, and the danger of overheating dairy.
- Make a simple pan crepe and pancake batter.
- Plate a clean breakfast to order.

Watch alongside this module (free)

[Jacques Pepin - The Secret to Perfectly Cooked Eggs \(KQED\)](#)

[Jacques Pepin - Techniques & More \(KQED\)](#)

[Gordon Ramsay \(knife & cooking basics\)](#)

Lesson 5.1 - Understanding the Egg

Objective: Know how egg proteins behave so you can control texture.

An egg is a protein toolkit. Whites set around 150 to 160 deg F, yolks a bit higher; push past that and proteins squeeze out water and turn rubbery. Almost every egg fault - rubbery scramble, chalky yolk, weeping custard - is overheating. The skill is gentle, controlled heat and stopping early.

Eggs also do work beyond breakfast: they leaven (whipped whites), thicken (custards), emulsify (mayonnaise, hollandaise), and bind (meatballs, breading). Fresh eggs hold together better for poaching; older whites are easier to whip. Bring eggs toward room temperature for even cooking and better volume.

Key points

- Whites set ~150-160 deg F; overheating makes eggs rubbery and weepy.
- Most egg failures are simply too much heat - go gentle and stop early.
- Eggs leaven, thicken, emulsify, and bind across all of cooking.
- Fresh eggs poach best; older whites whip to greater volume.

Common mistakes to avoid

- Cooking eggs over high heat 'to save time'.
- Walking away - eggs go from perfect to overcooked in seconds.
- Salting whipped whites early, which slows the foam.

PRACTICE

Cook two eggs the same way, one over high heat and one over low. Taste the difference that heat alone makes.

Lesson 5.2 - Scrambles and Omelets

Objective: Make soft scrambles and a clean rolled omelet.

French-style scrambled eggs are cooked low and slow with constant stirring to make small, creamy curds; American-style use higher heat and bigger folds. Either way, pull them off the heat while still slightly underdone - carryover finishes them - and season at the end.

The rolled French omelet is the benchmark: beaten eggs poured into a hot buttered nonstick pan, stirred briefly, then rolled into a pale, tender cylinder with no browning. It takes practice and a relaxed wrist. The diner omelet, by contrast, is set flatter and folded over a filling.

Key points

- Low heat plus constant stirring = small, creamy curds.
- Pull eggs off heat slightly underdone; carryover finishes them.
- A French omelet is pale and tender, with no browning.
- Season at the end so salt does not toughen the eggs early.

Technique, step by step

1	Beat	Whisk eggs until uniform; no streaks of white.
2	Heat gently	Melt butter in a nonstick pan over low-medium heat.
3	Stir	Stir constantly for creamy curds, easing off as they set.
4	Roll	Tilt the pan and roll the omelet onto the plate, seam down.

Common mistakes to avoid

- High heat that browns and toughens the eggs.
- Cooking until fully set in the pan, so they overcook on the plate.
- Using a pan that sticks - reach for nonstick here.

PRACTICE

Make a rolled French omelet with no browning. Repeat until it comes out pale, tender, and cleanly rolled.

Lesson 5.3 - Poached, Fried, and Boiled Eggs

Objective: Cook eggs to order: runny poach, crisp fry, jammy boil.

Poaching: use the freshest eggs, barely simmering water with a splash of vinegar, slip the egg into a gentle swirl, and cook three to four minutes for set whites and a runny yolk. Fried eggs range from sunny-side-up (gentle, covered) to over-easy (flipped) to crispy (hotter fat, lacy edges).

Boiled eggs are all about timing from the same starting point. From boiling water: about 6 to 7 minutes gives a jammy yolk, 9 to 12 minutes a fully set yolk. Shock in ice water to stop the cook and make peeling easier. Slightly older eggs peel more cleanly than very fresh ones.

Key points

- Poach fresh eggs in barely-simmering water; 3-4 minutes for a runny yolk.
- Fried eggs: low and covered for sunny-side-up; hotter for crispy edges.
- Boiled: ~6-7 min jammy, ~9-12 min fully set, then ice-shock.
- Ice-shocking stops carryover and eases peeling.

Technique, step by step

1	Swirl	Create a gentle vortex in barely-simmering water.
2	Slip	Lower the egg in close to the water; let the white wrap the yolk.
3	Time	Cook 3-4 minutes for a runny center.
4	Lift	Remove with a slotted spoon and drain on a towel.

Common mistakes to avoid

- Boiling the poaching water hard, which shreds the white.
- Using old eggs for poaching, so the white disperses.
- Skipping the ice bath, so boiled eggs overcook and stick to the shell.

PRACTICE

Poach two eggs with set whites and runny yolks. Separately, boil eggs to a jammy center and ice-shock them.

Lesson 5.4 - Custards and the Griddle

Objective: Make a smooth custard and cook crepes and pancakes.

Custards are thickened with eggs and gentle heat. Creme anglaise - a pourable vanilla custard - is made by tempering: whisk a little hot milk into the yolks to warm them gradually, return everything to low heat, and stir until it coats the back of a spoon (nappe), around 170 to 180 deg F. Strain and chill. Push too far and it scrambles.

Griddle work is about heat and patience. Crepes use a thin batter swirled into a hot pan and flipped once. Pancakes use a thicker, leavened batter; cook until bubbles form and the edges set, then flip a single time. A properly preheated, lightly greased surface gives even color.

Key points

- Temper eggs - add hot liquid slowly - to avoid scrambling.
- Cook custard to nappe (~170-180 deg F): it coats the back of a spoon.
- Strain custards for a silky texture; chill promptly.
- Pancakes: flip once when bubbles form and edges set.

Technique, step by step

1	Temper	Whisk a little hot milk into the yolks to warm them.
2	Return	Pour back into the pot over low heat.

- | | | |
|---|-------------------------|---|
| 3 | Stir to nappe | Stir constantly until it coats a spoon. |
| 4 | Strain and chill | Pass through a sieve and cool quickly. |

Common mistakes to avoid

- Overheating custard so it scrambles into sweet egg.
- Flipping pancakes more than once and deflating them.
- Cooking crepes on a cold pan so they steam and tear.

PRACTICE

Make creme anglaise without scrambling it - temper, stir to nappe, strain, chill. Then make a short stack of pancakes flipping once.

MODULE 06 - Weeks 10-11

Vegetables, Grains & Legumes

Big idea: Treat vegetables, grains, and legumes with the same rigor as protein and your whole cooking improves.

The foundation of flavor, color, and a modern plate. Treat plants with the same rigor as protein.

What you'll be able to do

- Blanch, shock, glaze, roast, and puree vegetables.
- Cook rice, grains, and legumes to correct texture.
- Build a balanced vegetarian plate.
- Preserve color, nutrients, and texture.

Watch alongside this module (free)

[Ethan Chlebowski \(the 'why' behind cooking\)](#)

[Pro Home Cooks \(skill-building series\)](#)

[J. Kenji Lopez-Alt \(food science & technique\)](#)

Lesson 6.1 - Blanching, Shocking, and Boiling

Objective: Cook vegetables in water while keeping color and texture.

Blanching means briefly boiling a vegetable in well-salted water, then shocking it in ice water to stop the cook. The result is a green vegetable that is vivid, crisp-tender, and seasoned through - far better than one boiled to gray mush. Blanching also loosens skins (tomatoes) and tames raw bites.

Salt the water generously - it should taste like the sea - because this is your one chance to season vegetables from within. Cook green vegetables uncovered to let acids escape and protect color, and do not overcrowd the pot, which drops the temperature and stretches the cook time.

Key points

- Blanch in heavily salted boiling water, then ice-shock to set color.
- Salt the water 'like the sea' to season vegetables from within.
- Cook green vegetables uncovered to preserve their color.
- Do not crowd the pot; work in batches to keep the water hot.

Technique, step by step

1	Boil	Bring a big pot of heavily salted water to a rolling boil.
2	Blanch	Add the vegetable and cook until just crisp-tender.
3	Shock	Plunge into ice water to stop cooking and lock in color.
4	Drain	Drain well before saucing or storing.

Common mistakes to avoid

- Under-salting the water, leaving vegetables bland.
- Skipping the ice bath, so carryover dulls the color.
- Crowding the pot and steaming instead of blanching.

PRACTICE

Blanch and shock a green vegetable, then cook a second batch to death by boiling. Compare color, texture, and flavor.

Lesson 6.2 - Roasting, Sauteing, and Glazing

Objective: Use dry heat and glazing to concentrate vegetable flavor.

Dry heat caramelizes vegetable sugars and concentrates flavor. Roast vegetables in a hot oven (400 to 450 deg F) on a single uncrowded layer with enough oil and salt; crowding traps steam and prevents browning. Sauteing does the same on the stovetop for quicker-cooking vegetables.

Glazing finishes root vegetables in a little stock, butter, and sugar until the liquid reduces to a shiny coating. The technique builds a glossy, lightly sweet exterior while the vegetable cooks through. Across all of these, dryness and space on the pan are what let vegetables brown instead of steam.

Key points

- Roast hot (400-450 deg F) in a single, uncrowded layer.
- Use enough oil and salt; dry surfaces brown, wet ones steam.
- Saute quicker vegetables with movement and high heat.
- Glaze roots in stock, butter, and a little sugar to a shiny coat.

Common mistakes to avoid

- Crowding the sheet pan so vegetables steam and stay pale.
- Skimping on oil or salt, so nothing caramelizes.
- Stirring roasting vegetables constantly, preventing a crust.

PRACTICE

Roast a tray of vegetables in a single layer until deeply browned. Glaze a pan of carrots in stock and butter until shiny.

Lesson 6.3 - Rice and Grains

Objective: Cook rice and grains to the right texture, including risotto.

Most grains cook by absorption: a set ratio of grain to liquid simmered covered until the liquid is gone, then rested off heat. White rice is roughly 1 to 1.5 or 2 parts water; the pilaf method first toasts the grain in fat for nuttier, separate grains. Different grains want different ratios, so learn them by weight.

Risotto is the exception that proves the rule: short-grain rice cooked with warm stock added a ladle at a time, stirred so the grains release starch and turn creamy without any cream. Finish off heat with butter and cheese (mantecatura) for gloss. The goal is loose and flowing, not stiff.

Key points

- Most grains cook by absorption at a fixed liquid ratio, then rest.
- Pilaf method: toast the grain in fat first for separate, nutty grains.
- Risotto gets creamy from stirring out starch, not from cream.
- Finish risotto off heat with butter and cheese for gloss.

Technique, step by step

1	Toast	For pilaf/risotto, toast the grain in fat until fragrant.
2	Add liquid	Absorption: add it all; risotto: a ladle at a time.
3	Cook	Simmer covered (absorption) or stir constantly (risotto).
4	Rest or finish	Rest absorption grains; mantecare risotto with butter and cheese.

Common mistakes to avoid

- Lifting the lid and stirring absorption rice, making it gummy.
- Adding cold stock to risotto, which stalls the cook.
- Cooking risotto until stiff instead of loose and flowing.

PRACTICE

Cook a risotto, adding warm stock a ladle at a time and stirring to creaminess, then finish off heat with butter and cheese.

Lesson 6.4 - Legumes and Purees

Objective: Cook beans and lentils well and make smooth purees.

Dried legumes reward a little planning. Soaking beans shortens cooking and evens the texture; lentils need no soak. Cook them gently in plenty of water with aromatics, and salt them - contrary to old myth, salting during cooking seasons them through without keeping them hard. Cook until fully creamy inside, never chalky.

Purees turn cooked vegetables or legumes into smooth, sauce-like textures. Cook the ingredient soft, blend with some of its liquid and fat, and pass through a sieve for silkiness. Season aggressively - blending dulls salt perception. A good puree is a plate component, a sauce, and a dip all at once.

Key points

- Soak most dried beans to even the cook; lentils need no soak.
- Salt legumes while cooking - it seasons through and does not toughen them.
- Cook beans until fully creamy inside, not chalky.
- Blend purees with their liquid and fat; season boldly and strain.

Common mistakes to avoid

- Boiling beans hard, which bursts skins and makes them mushy outside, raw inside.
- Under-seasoning purees, which taste flat after blending.
- Adding acid (tomato, vinegar) early, which can keep beans firm.

PRACTICE

Cook a pot of dried beans with aromatics and salt until creamy. Blend half into a smooth, well-seasoned puree and strain it.

MODULE 07 - Weeks 12-13

Proteins I: Poultry & Meat

Big idea: Butchery basics, the big roasting and braising methods, and judging meat by temperature.

Butchery basics, the big roasting and braising methods, and judging doneness by temperature.

What you'll be able to do

- Break down a whole chicken into eight pieces.
- Roast, sear, and braise meat and poultry to target temperatures.
- Understand cuts: which are for fast heat vs. low-and-slow.
- Rest meat and build a pan sauce or jus.

Watch alongside this module (free)

[Jacques Pepin - Classic Roast Chicken \(KQED\)](#)

[J. Kenji Lopez-Alt \(food science & technique\)](#)

[Brian Lagerstrom \(ex-restaurant pro, approachable\)](#)

Lesson 7.1 - Meat Science and Buying

Objective: Understand muscle, fat, and collagen so you can match cut to method.

Meat is muscle, fat, and connective tissue. Hard-working muscles (shoulder, shank, leg) are full of collagen and flavor but tough - they need low, slow, moist cooking to melt that collagen into gelatin. Tender muscles that do little work (loin, rib) suit fast, dry, high heat. Match the cut to the method and you rarely go wrong.

Buy with your eyes and nose: meat should look fresh and smell clean, with fat that suits the cut. Marbling - flecks of intramuscular fat - means flavor and juiciness. A relationship with a butcher gets you better cuts and free bones for stock. Let meat come toward room temperature before cooking for even results.

Key points

- Tough, hard-working cuts (chuck, shank) want low-and-slow moist heat.
- Tender cuts (loin, rib) want fast, dry, high heat.
- Marbling = intramuscular fat = flavor and juiciness.
- Let meat temper toward room temp before cooking for even doneness.

Common mistakes to avoid

- Grilling a tough braising cut, leaving it chewy.
- Slow-braising a tender steak cut, drying it out.
- Cooking ice-cold meat, so the outside overcooks before the center warms.

PRACTICE

Look at the cuts in a market and sort them in your head into 'fast/dry' and 'slow/moist'. Buy one of each for the next two lessons.

Lesson 7.2 - Breaking Down a Whole Chicken

Objective: Fabricate a whole chicken into eight pieces and save the carcass.

Breaking down a chicken is cheaper than buying parts, gives you a carcass for stock, and teaches you how the animal goes together. The eight-piece cut yields two drumsticks, two thighs, two wings, and two breasts. Work with the joints, not against them - find where bones meet and the knife slides through with little force.

Remove the legs first by cutting through the skin and popping the hip joint, separate drumstick from thigh at the knee, take off the wings at the shoulder, then split the breast off the carcass along the keel bone. A boning or chef's knife and confident, deliberate cuts are all you need.

Key points

- The eight-piece cut: 2 drumsticks, 2 thighs, 2 wings, 2 breasts.
- Cut through joints, not bone - find the gap and the knife glides.
- Keep the carcass and trim for stock (Module 4).
- A sharp boning or chef's knife and deliberate cuts do the job.

Technique, step by step

1	Legs	Cut the skin, bend the leg back to pop the hip, and cut through.
2	Split	Separate drumstick from thigh at the knee joint.
3	Wings	Remove each wing at the shoulder joint.
4	Breasts	Cut along the keel bone to free each breast from the carcass.

Common mistakes to avoid

- Hacking through bone instead of finding the joint.
- Tearing the skin by rushing - let the knife do the work.
- Throwing away the back and trim that make great stock.

PRACTICE

Break down one whole chicken into eight pieces. Save the carcass and make stock with it.

Lesson 7.3 - Roasting and Searing

Objective: Roast poultry and sear steaks to a target temperature.

Roasting cooks with dry oven heat. For a whole chicken, dry the skin well, season generously, and roast hot enough to crisp the skin while the meat reaches 165 deg F in the breast. Trussing helps it cook evenly. Rest before carving so juices settle. The same logic scales to larger roasts with carryover in mind.

Searing steaks is about a hot pan and patience: pat dry, salt, lay in a preheated pan with a little oil, and leave it until a crust forms. The reverse sear - low oven first, then a hard sear - gives edge-to-edge doneness and a great crust on thick steaks. Always finish by temperature, then rest.

Key points

- Dry the skin and season well for crisp roast chicken; breast to 165 deg F.
- Sear in a preheated pan and do not move the meat until it crusts.
- Reverse sear thick steaks: gentle oven, then a hard finishing sear.
- Cook to temperature, then rest before slicing.

Technique, step by step

1	Dry and season	Pat dry, salt ahead, and let the surface dry.
2	Roast or sear	High oven for poultry; preheated pan for steaks.
3	Check temp	Pull a few degrees early for carryover.
4	Rest	Rest before carving or slicing to keep juices in.

Common mistakes to avoid

- Roasting damp-skinned chicken, which steams instead of crisping.
- Moving the steak around so it never builds a crust.
- Slicing immediately and losing the juices.

PRACTICE

Roast a chicken to 165 deg F in the breast and rest it. Separately, sear a steak to medium-rare by temperature and rest it.

Lesson 7.4 - Braising Tough Cuts

Objective: Turn a cheap, tough cut into something tender and rich.

Braising is the magic trick of cheap cooking: take a tough, collagen-rich cut, brown it hard for flavor, then cook it low and slow partly submerged in liquid until the collagen melts into gelatin and the meat becomes fork-tender. The braising liquid, enriched by everything that cooked in it, becomes the sauce.

The method is forgiving but has rules: sear well first, use only enough liquid to come partway up the meat, keep it at a bare simmer (the oven is ideal for even heat), and give it time. When the meat yields to a fork, strain and reduce the liquid into a glossy sauce, skimming excess fat.

Key points

- Sear hard first - that browning is most of the flavor.
- Add liquid only partway up the meat; full submersion boils, not braises.
- Keep a bare simmer (a low oven gives steady, even heat).
- Reduce the strained braising liquid into the sauce.

Technique, step by step

1	Sear	Brown the meat deeply on all sides; remove.
2	Build	Soften aromatics, deglaze, return the meat with partial liquid.
3	Braise	Cover and cook low until fork-tender.

4**Finish**

Skim, strain, and reduce the liquid to a sauce.

Common mistakes to avoid

- Skipping the sear and getting a pale, flat-tasting braise.
- Boiling it hard, which makes the meat dry and stringy.
- Pulling it early while the collagen is still tough.

PRACTICE

Braise a tough cut (chuck, short rib, or shank) until fork-tender, then reduce the braising liquid into a sauce.

MODULE 08 - Week 14

Proteins II: Fish & Seafood

Big idea: Seafood is delicate, fast, and unforgiving of overcooking; learn to buy, store, and cook it with confidence.

Delicate, fast, and unforgiving of overcooking. Learn to buy, store, and cook seafood with confidence.

What you'll be able to do

- Judge freshness and store seafood correctly.
- Pan-sear fish for crisp skin; poach and steam delicate fillets.
- Cook shellfish (shrimp, mussels, scallops) to just-done.
- Portion a fillet and remove pin bones.

Watch alongside this module (free)

[America's Test Kitchen \(tested technique\)](#)

[Ethan Chlebowski \(the 'why' behind cooking\)](#)

[ChefSteps \(modern technique & sous vide\)](#)

Lesson 8.1 - Buying and Storing Seafood

Objective: Choose fresh seafood and store it correctly.

Seafood spoils faster than meat, so freshness is everything. Fresh fish smells clean and briny, never fishy or sour; eyes are clear, flesh is firm and springs back, and fillets look moist and translucent, not dull or gaping. If you cannot get truly fresh, well-handled frozen-at-sea fish is often a better choice.

Store seafood cold - ideally on ice in the fridge - and cook it within a day or two. Keep it dry and well-wrapped. Plan your cooking around when you can buy it fresh, not the other way around. Sustainability guides can help you choose species that are well managed.

Key points

- Fresh fish smells briny and clean, never sour; flesh is firm and springy.
- Frozen-at-sea fish can beat tired 'fresh' fish at the counter.
- Store cold, dry, and well-wrapped; cook within a day or two.
- Buy when it is freshest and plan the meal around it.

Common mistakes to avoid

- Buying fish that smells strongly fishy or looks dull and gaping.
- Letting seafood sit warm or swim in its own liquid.
- Holding fresh fish for days before cooking it.

PRACTICE

At a fish counter, practice judging freshness by smell, eyes, and firmness. Buy the freshest fillet and cook it the same day.

Lesson 8.2 - Pan-Searing Fish for Crisp Skin

Objective: Get crisp skin and a moist interior on a fillet.

Crisp skin comes from dryness, heat, and patience. Pat a skin-on fillet very dry, season it, and lay it skin-side down into a preheated pan with oil. Then press gently for a few seconds so the skin makes full contact and does not curl. Leave it alone - most of the cooking happens on the skin side.

Cook until you can see the flesh turning opaque most of the way up, then flip for just a short finish on the second side. Fish keeps cooking off heat, so pull it a touch early. The flesh should flake but still look juicy; opaque and just-firm is done, dry and chalky is overdone.

Key points

- Dry skin + preheated pan + a brief press = crisp, flat skin.
- Cook mostly on the skin side; flip only to finish.
- Pull fish a little early - it cooks on off heat.
- Done = opaque and just-firm, flaking but still juicy.

Technique, step by step

1	Dry and season	Pat the fillet bone-dry; salt the skin.
2	Lay skin-down	Place into a hot, oiled pan, skin-side down.
3	Press	Press gently for a few seconds so the skin stays flat.
4	Flip late	Cook through on the skin, then flip briefly to finish.

Common mistakes to avoid

- Wet skin that steams and sticks instead of crisping.
- Flipping too early and too often, so the skin never crisps.
- Overcooking until the flesh is dry and chalky.

PRACTICE

Pan-sear a skin-on fillet to crisp skin without tearing it, cooking mostly on the skin side and flipping only to finish.

Lesson 8.3 - Poaching and Steaming

Objective: Cook delicate fish gently in liquid or vapor.

Gentle, moist methods suit lean and delicate fish. Poaching holds fish in barely-simmering liquid - water, court bouillon, milk, or oil - at low temperature so the flesh stays silky and moist. The liquid can be flavored with aromatics and later become a sauce. Keep the heat low; a hard boil toughens and breaks the fish.

Steaming cooks fish in vapor over simmering liquid, often with aromatics like ginger, scallion, and herbs. It is clean, fast, and forgiving, and it preserves the fish's natural texture. Both methods are nearly foolproof if you respect the gentle heat and stop as soon as the flesh turns opaque.

Key points

- Poach in barely-simmering liquid; never let it boil hard.
- Court bouillon (aromatic poaching liquid) adds flavor and becomes a sauce.
- Steaming preserves texture and takes well to aromatics.
- Stop as soon as the flesh turns opaque.

Common mistakes to avoid

- Boiling the poaching liquid, which toughens and shreds the fish.
- Over-steaming until the fish is dry.
- Crowding the steamer so it cooks unevenly.

PRACTICE

Poach a fillet in gently simmering aromatic liquid, or steam one over ginger and scallion, pulling it the moment it turns opaque.

Lesson 8.4 - Shellfish

Objective: Cook shrimp, mussels, and scallops to just-done.

Shellfish cook fast and punish overcooking with rubberiness. Shrimp are done the moment they curl into a loose C and turn opaque - a tight O means overdone. Mussels and clams steam open in minutes in a little aromatic liquid; discard any that stay shut, and use the flavorful broth they release as a sauce.

Scallops are a searing showcase: pat them very dry, season, and sear hard in a hot pan to build a golden crust while the center stays barely warm and translucent. Across all shellfish, the rule is the same - cook just to done and pull immediately, because carryover finishes the job.

Key points

- Shrimp: a loose C and opaque flesh is done; a tight O is overcooked.
- Mussels/clams steam open in minutes; discard any that stay shut.
- Use the broth shellfish release as an instant sauce.
- Scallops: bone-dry, hot pan, hard sear, barely-warm center.

Technique, step by step

1	Dry	Pat shellfish very dry for searing; rinse and debeard mussels.
2	High heat	Use a hot pan for scallops; a covered pot for mussels.
3	Watch closely	Pull shrimp/scallops at just-done; mussels when open.
4	Sauce it	Build a quick pan broth from the released liquid.

Common mistakes to avoid

- Cooking shrimp to a tight, rubbery O.
- Searing wet scallops so they steam instead of browning.
- Eating mussels that never opened.

PRACTICE

Steam mussels and build a quick pan broth to serve with bread. Separately, sear scallops with a golden crust and a barely-warm center.

MODULE 09 - Weeks 15-16

Baking & Pastry Fundamentals

Big idea: Baking is chemistry with deadlines; precision, ratios, and a scale replace improvisation.

Baking is chemistry with deadlines. Precision, ratios, and a scale replace improvisation.

What you'll be able to do

- Work by weight; understand baker's percentages.
- Make a basic bread dough and understand gluten and fermentation.
- Produce pie dough, a simple cake, and a custard dessert.
- Understand leavening: yeast, chemical, and mechanical.

Watch alongside this module (free)

[Brian Lagerstrom \(ex-restaurant pro, approachable\)](#)

[America's Test Kitchen \(tested technique\)](#)

[ChefSteps \(modern technique & sous vide\)](#)

Lesson 9.1 - Precision and Baker's Percentages

Objective: Work by weight and understand baker's percentages.

Baking is less forgiving than savory cooking because structure depends on exact ratios. The fix is to weigh everything in grams - a cup of flour can vary by 30 percent depending on how it is scooped, and that ruins recipes. A scale makes baking repeatable and lets you scale recipes up or down with confidence.

Baker's percentage expresses every ingredient as a percent of the flour weight (flour is always 100 percent). A bread at 65 percent hydration has 65 grams of water per 100 grams of flour. This shared language lets you compare and adjust formulas precisely instead of guessing.

Key points

- Weigh ingredients in grams; volume measures are unreliable for baking.
- Flour = 100 percent; every other ingredient is a percent of flour weight.
- Hydration = water as a percent of flour; it controls dough texture.
- A scale makes recipes repeatable and easy to scale.

Common mistakes to avoid

- Measuring flour by the cup and getting wildly different amounts.
- Eyeballing leaveners and salt, which need precision.
- Ignoring ingredient temperature, which affects doughs and batters.

PRACTICE

Reweigh a baking recipe in grams and convert it to baker's percentages. Bake it by weight and note how consistent it is.

Lesson 9.2 - Bread and Gluten

Objective: Make a basic dough and understand gluten and fermentation.

Bread is flour, water, salt, and leavening transformed by time. Mixing flour and water develops gluten - a stretchy protein network that traps gas and gives bread its chew and structure. Kneading (or folding and resting) builds that network; a windowpane of dough you can stretch thin without tearing signals it is developed.

Fermentation is where flavor lives. Yeast (or a sourdough starter) produces gas that leavens the dough and acids that build taste; slower, cooler fermentation deepens flavor. After shaping and a final proof, a hot oven sets the crust and crumb. Patience, not force, makes good bread.

Key points

- Gluten is the stretchy protein network that gives bread structure.
- Develop it by kneading or by fold-and-rest cycles.
- Fermentation builds flavor; slower and cooler is deeper.
- Proof until puffy and jiggy, then bake hot to set crust and crumb.

Technique, step by step

1	Mix	Combine flour, water, salt, and yeast into a shaggy dough.
2	Develop	Knead or do stretch-and-folds to build gluten.
3	Ferment	Let it rise until roughly doubled and airy.
4	Shape and bake	Shape, proof, and bake in a hot oven.

Common mistakes to avoid

- Adding too much flour, making a dense, dry loaf.
- Rushing fermentation, so the bread is bland and tight.
- Baking in a cool oven, giving a pale, heavy crust.

PRACTICE

Bake a simple loaf by weight. Log the hydration and the crumb, then adjust next time based on what you saw.

Lesson 9.3 - Pie and Tart Dough

Objective: Make a flaky, all-butter dough and blind-bake a shell.

Flaky pastry depends on keeping the fat cold and in pieces. When cold butter is cut into flour and then hit with oven heat, its water turns to steam and pushes the dough into layers. Work fast and cold, add just enough ice water to bring the dough together, and do not overwork it or you build gluten and toughen the crust.

The classic 3-2-1 ratio - three parts flour, two parts fat, one part water by weight - gives a reliable dough. Rest it cold to relax the gluten and firm the butter, roll it evenly, and chill the shaped shell before baking.

Blind-baking with weights keeps the bottom flat and crisp for wet fillings.

Key points

- Keep the butter cold and in pieces for flaky layers.
- 3-2-1 by weight: flour, fat, water.
- Add minimal water and do not overwork the dough.
- Rest cold before rolling; blind-bake shells for wet fillings.

Technique, step by step

1	Cut in fat	Cut cold butter into flour until pea-sized pieces remain.
2	Add water	Sprinkle in just enough ice water to bring it together.
3	Rest	Chill the disc to relax gluten and firm the butter.
4	Roll and chill	Roll evenly, shape, chill, then blind-bake if needed.

Common mistakes to avoid

- Warm butter that blends in, giving a dense, greasy crust.
- Too much water and kneading, toughening the dough.
- Skipping the chill, so the shell slumps in the oven.

PRACTICE

Make an all-butter pie dough by the 3-2-1 ratio and blind-bake a shell that stays flat and crisp.

Lesson 9.4 - Cakes and Custards

Objective: Bake a tender cake and set a simple custard dessert.

Cakes rise by trapping air and gas in a batter that then sets. The creaming method beats sugar into butter to whip in air, then adds eggs and alternating dry and wet ingredients; the foaming method whips eggs for lift. Do not overmix once the flour goes in - that builds gluten and toughens the crumb. Bake until a tester comes out clean.

Baked custards - from creme caramel to a simple pot de creme - set with gentle, even heat, often in a water bath that buffers the temperature so the edges do not overcook before the center sets. A custard is done when it is set at the edges but still jiggles slightly in the middle; carryover finishes it.

Key points

- Creaming whips air into butter and sugar for a tender crumb.
- Do not overmix after adding flour, or the cake turns tough.
- Bake custards gently, often in a water bath, for even setting.
- Custard is done when set at the edges but still jiggly in the center.

Common mistakes to avoid

- Overmixing batter and getting a dense, rubbery cake.
- Baking custard too hot, so it curdles and weeps.
- Opening the oven early and collapsing a delicate cake.

PRACTICE

Bake one cake by the creaming method and evaluate the crumb. Set a simple baked custard in a water bath to a gentle jiggle.

MODULE 10 - Weeks 17-19

Global Cuisines: French, Italian, Asian & Latin

Big idea: Apply your fundamentals across four major traditions and learn the pantry and techniques of each.

Apply your fundamentals across four major traditions and learn the pantry and techniques of each.

What you'll be able to do

- Cook a fresh pasta and an emulsified pasta sauce (Italian).
- Build a stir-fry with wok hei principles (Asian).
- Make a salsa, a braise, and handle masa basics (Latin).
- Execute a classic French bistro plate end to end.

Watch alongside this module (free)

[J. Kenji Lopez-Alt \(food science & technique\)](#)

[Ethan Chlebowski \(the 'why' behind cooking\)](#)

[Pro Home Cooks \(skill-building series\)](#)

[Brian Lagerstrom \(ex-restaurant pro, approachable\)](#)

Lesson 10.1 - Italian: Pasta and Sauce

Objective: Make fresh pasta and emulsify a pasta sauce properly.

Italian cooking prizes a few great ingredients handled simply. Fresh egg pasta is flour and eggs kneaded, rested, rolled thin, and cut; the dough should be smooth and elastic. Dried pasta is excellent too - the key skill is cooking it in well-salted water to al dente, firm to the bite, because it finishes in the sauce.

The secret to a glossy Italian sauce is emulsion: finish the pasta in the pan with the sauce and a splash of starchy pasta water, tossing hard so the fat and water bind into a silky coat that clings to every strand. Cacio e pepe and aglio e olio live or die on this move. Sauce the pasta; do not drown it.

Key points

- Cook pasta in well-salted water to al dente - it finishes in the sauce.
- Save starchy pasta water; it is the key to emulsified sauce.
- Toss pasta and sauce hard in the pan to bind fat and water.
- Sauce should coat the pasta, not pool around it.

Technique, step by step

1	Boil	Cook pasta in salted water until just shy of al dente.
2	Combine	Move pasta into the pan with the sauce and some pasta water.
3	Emulsify	Toss vigorously over heat until glossy and clinging.
4	Finish	Adjust with more water, cheese, or fat to a silky coat.

Common mistakes to avoid

- Rinsing pasta and washing away the starch that builds sauce.
- Drowning pasta in watery sauce that will not cling.
- Over-flouring fresh dough so it turns dry and tough.

PRACTICE

Make fresh pasta by hand, or cook good dried pasta, and finish it in the pan with an emulsified butter or tomato sauce and pasta water.

Lesson 10.2 - Asian: The Stir-Fry

Objective: Cook a fast, high-heat stir-fry with proper prep.

Stir-frying is fast cooking over very high heat, so everything depends on mise en place: all ingredients cut to even, quick-cooking sizes and arranged in cooking order before the wok ever heats. Once you start, there is no time to chop. Cook in stages so nothing crowds and steams, then bring it together at the end.

The flavor framework across much of East and Southeast Asia balances salty, sweet, sour, and umami - soy or fish sauce, a touch of sugar, vinegar or citrus, and aromatics like garlic, ginger, and chili. Get the pan screaming hot, keep things moving, and build a quick sauce that coats without drowning.

Key points

- Prep everything first - stir-frying is too fast to chop mid-cook.
- Use very high heat and cook in stages to avoid steaming.
- Balance salty, sweet, sour, and umami in the sauce.
- Aromatics (garlic, ginger, chili) go in fast and must not burn.

Technique, step by step

1	Prep	Cut everything to even sizes; line up in cooking order.
2	Heat	Get the wok or pan very hot before adding oil.
3	Cook in stages	Sear proteins, then vegetables, keeping things moving.
4	Sauce	Add the balanced sauce at the end and toss to coat.

Common mistakes to avoid

- Starting to cook before all the prep is done.
- Crowding the pan so the food steams and turns gray.
- Burning the garlic and ginger over too-high heat.

PRACTICE

Cook a high-heat stir-fry with fully prepped mise en place and a sauce balanced across salty, sweet, sour, and umami.

Lesson 10.3 - Latin: Salsas and Braises

Objective: Build fresh salsas and a flavorful braise.

Latin American cooking spans a huge range, but two techniques travel everywhere: the fresh salsa and the slow braise. Salsas - raw or charred - layer chiles, acid, alliums, and herbs into bright, punchy condiments that wake up almost any plate. Toasting or charring the components deepens their flavor before blending or chopping.

Braises like a chile-stewed pork or beef build deep flavor from a base of toasted chiles, aromatics, and spices, cooked low and slow until the meat shreds. Beans, rice, and masa (for tortillas and tamales) round out the table. The throughline is layered flavor built patiently from humble ingredients.

Key points

- Salsas layer chiles, acid, alliums, and herbs for brightness.
- Toasting or charring components deepens salsa and braise flavor.
- Latin braises build from toasted chiles and spices, cooked low and slow.
- Beans, rice, and masa complete the plate.

Common mistakes to avoid

- Skipping the toast/char step, leaving flavors flat.
- Under-acidifying a salsa, so it tastes dull.
- Rushing a braise before the meat is shreddable.

PRACTICE

Prepare one Latin braise plus a fresh salsa to accompany it, toasting or charring the salsa components for depth.

Lesson 10.4 - French Bistro Classics

Objective: Execute a classic French plate that ties earlier modules together.

French bistro cooking is where your fundamentals converge: a seared protein, a pan sauce from fond, a vegetable cooked with care, and a starch - all the techniques from earlier modules on one plate. Dishes like steak frites, roast chicken with jus, or fish meuniere are simple in concept and rigorous in execution.

The discipline is in the details: properly seasoned and rested meat, a balanced sauce, a vegetable that keeps its color, and clean plating. Nothing here is exotic; it is fundamentals done precisely. Master a few bistro plates and you have a repertoire you can cook for anyone, anytime.

Key points

- Bistro plates combine a seared protein, a pan sauce, a vegetable, and a starch.
- It is fundamentals done precisely, not exotic technique.
- Season, rest, and sauce with care; keep vegetables vivid.
- A few mastered bistro dishes form a lifelong repertoire.

Common mistakes to avoid

- Treating 'simple' as 'careless' - bistro food exposes sloppy technique.
- A pan sauce that is unbalanced or greasy.

- Overcooked, dull vegetables next to a good protein.

PRACTICE

Cook a complete French bistro plate - seared protein, pan sauce from fond, a carefully cooked vegetable, and a starch.

MODULE 11 - Weeks 20-21

Flavor, Seasoning & Menu Development

Big idea: Move from following recipes to composing dishes by learning to taste, balance, and design.

Move from following recipes to composing dishes. Learn to taste, balance, and design a menu.

What you'll be able to do

- Season confidently with salt, acid, fat, and heat.
- Diagnose and fix an under-seasoned or unbalanced dish.
- Build flavor with aromatics, herbs, spices, and umami.
- Design a small, coherent multi-course menu.

Watch alongside this module (free)

[Ethan Chlebowski \(the 'why' behind cooking\)](#)

[Adam Ragusea \(practical home technique\)](#)

[Internet Shaquille \(efficiency & fundamentals\)](#)

Lesson 11.1 - Salt, Fat, Acid, Heat

Objective: Use the four levers that make food taste good.

Most of what separates flat food from delicious food comes down to four elements. Salt amplifies flavor and suppresses bitterness; most home food is under-salted. Fat carries flavor and creates richness and texture. Acid - vinegar, citrus, wine - brightens and balances richness. Heat (both temperature and chili) shapes texture and adds another dimension.

Seasoning is not a single step at the end; it is a series of adjustments throughout cooking, tasting as you go. When a dish tastes 'off' but you cannot name why, run through the four: does it need salt, fat, acid, or heat? Usually one of them is the answer, and acid is the most commonly missing.

Key points

- Salt amplifies flavor; most home cooking is under-salted.
- Fat carries flavor and builds richness and texture.
- Acid brightens and balances; it is the most often missing element.
- Season throughout and taste constantly, not just at the end.

Common mistakes to avoid

- Salting only at the end instead of in layers.
- Forgetting acid, leaving rich dishes heavy and dull.
- Fearing salt and serving flat food.

PRACTICE

Take a bland dish and improve it using only salt, acid, fat, and heat, tasting after each change and noting what each one did.

Lesson 11.2 - Tasting and Adjusting

Objective: Build the habit of tasting and diagnosing a dish.

The most important tool a cook has is a spoon. Tasting constantly - and knowing what to do about what you taste - is the difference between cooking to a recipe and cooking with judgment. Taste at every stage, and learn to ask a single question: what does this need? Then make one change at a time and taste again.

Develop a vocabulary for what is wrong: flat (needs salt), heavy (needs acid), thin (needs fat or reduction), harsh (needs cooking time or sweetness), one-note (needs contrast or a finishing element). Adjusting is a loop - taste, diagnose, adjust, taste - that you run until the dish is balanced.

Key points

- Taste at every stage; the spoon is your most important tool.
- Ask 'what does this need?' and change one thing at a time.
- Flat = salt, heavy = acid, thin = fat/reduction, harsh = time/sweetness.
- Adjusting is a loop: taste, diagnose, adjust, taste again.

Common mistakes to avoid

- Not tasting until the food is plated.
- Changing several things at once and losing track of cause and effect.
- Over-correcting past the balance point.

PRACTICE

Cook a familiar dish and taste at every stage, writing down what you adjusted and why. Run the taste-diagnose-adjust loop deliberately.

Lesson 11.3 - Building Flavor Bases and Umami

Objective: Layer aromatics, spices, and umami for depth.

Deep flavor is built, not added at the end. Most cuisines start from an aromatic base - French mirepoix, Spanish and Latin sofrito, Italian soffritto, Cajun holy trinity - softened slowly in fat to create a foundation. Toasting whole spices and blooming ground spices in fat releases their oils and multiplies their impact.

Umami - the savory fifth taste - gives food its satisfying depth. It comes from glutamate-rich ingredients: tomato, mushrooms, aged cheese, soy sauce, fish sauce, cured meats, and stock. Layering a few umami sources, and finishing with fresh contrast (herbs, acid, crunch), is how a dish goes from fine to crave-able.

Key points

- Start from a slowly softened aromatic base (mirepoix, sofrito, trinity).
- Toast whole spices and bloom ground spices in fat for more flavor.
- Umami (tomato, mushroom, cheese, soy, fish sauce, stock) adds savory depth.
- Finish with fresh contrast - herbs, acid, crunch.

Common mistakes to avoid

- Rushing the aromatic base so it never builds flavor.
- Adding spices raw at the end instead of blooming them.

- Relying on salt alone for depth instead of umami.

PRACTICE

Build a dish from a slowly cooked aromatic base, bloom your spices, and layer at least two umami sources. Finish with something fresh.

Lesson 11.4 - Menu Development

Objective: Design a small, coherent multi-course menu.

Composing a menu is about balance and flow across the whole meal, not just one great dish. Vary richness, texture, temperature, and color from course to course so the meal builds and never tires the palate. A heavy main wants a bright, light starter; a rich dessert wants a clean, acidic counterpoint somewhere before it.

Practicality matters as much as artistry. Design so that not everything needs the oven or your hands at the last minute; build a prep list and a timeline so courses land calmly. Consider cost and season - the best menus use what is fresh and affordable - and write the menu so each dish makes the next one taste better.

Key points

- Vary richness, texture, temperature, and color across courses.
- Balance a heavy course with a lighter, brighter one.
- Design for logistics - not everything can finish at once.
- Use seasonal, affordable ingredients and write a prep timeline.

Common mistakes to avoid

- Three rich courses in a row that exhaust the palate.
- A menu where every dish needs the oven at 7:00 sharp.
- Ignoring season and cost, making the meal expensive and flat.

PRACTICE

Design a three-course menu with balance across the courses, plus a shopping list and a backward timeline so it lands calmly.

MODULE 12 - Weeks 22-24

The Professional Kitchen: Line, Plating & Career

Big idea: Bring it all together at professional pace: timing, plating, and a real path into the industry.

Bring it together at professional pace: timing multiple dishes, plating cleanly, and planning a path into the industry.

What you'll be able to do

- Run a 'service': cook multiple dishes to land together, hot and on time.
- Plate with composition, height, sauce, and garnish.
- Understand the brigade system and kitchen roles.
- Plan a realistic path: stages, apprenticeships, portfolio, certifications.

Watch alongside this module (free)

[Brian Lagerstrom \(ex-restaurant pro, approachable\)](#)

[ChefSteps \(modern technique & sous vide\)](#)

[Pro Home Cooks \(skill-building series\)](#)

Lesson 12.1 - Mise en Place at Scale and Timing

Objective: Cook multiple dishes so they finish together, hot and on time.

The leap to 'professional' is mostly about timing and organization, not fancier technique. A line cook can send dozens of plates because every component is prepped and arranged in advance, and because they cook backward from when the plate must go out. Scale your mise en place: prep all components, label, and arrange your station logically.

Build a timeline that works backward from serving time, identifying what can be made ahead, what holds, and what must be fired at the last minute. 'Firing' means starting the final cook on each component so they all peak together. The skill is sequencing several cooks at once - and it only comes from doing it.

Key points

- Prep and arrange all components before service begins.
- Cook backward from serving time; know what holds and what fires late.
- 'Firing' = starting final cooks so components peak together.
- Sequencing multiple dishes is a practiced skill, not a talent.

Technique, step by step

1	Prep	Make every component you can ahead; label and store.
2	Timeline	Write a backward schedule from serving time.
3	Set the station	Arrange tools and components for the firing order.
4	Fire	Start final cooks in sequence so everything lands together.

Common mistakes to avoid

- No timeline, so dishes finish at wildly different times.
- Trying to prep and cook simultaneously under pressure.
- Letting hot food sit and die while another component lags.

PRACTICE

Cook a three-course meal where every dish lands hot and on time, working from a written backward timeline.

Lesson 12.2 - Plating

Objective: Plate with composition, height, sauce, and garnish.

We eat with our eyes first. Good plating starts with a clear focal point - usually the protein - and uses negative space so the plate does not look cluttered. Height and layering create visual interest; a sauce is placed with intention (pooled, streaked, or dotted), not poured over everything; and garnish must be edible and relevant, never decoration for its own sake.

Keep it clean and purposeful: wipe drips, vary texture and color, and make sure every element on the plate earns its place and tastes good with the others. Odd numbers and asymmetry often look more natural than rigid symmetry. The food should look like itself, only its best self.

Key points

- Establish a focal point and use negative space.
- Build height and layers; place sauce with intention.
- Garnish must be edible and relevant, not mere decoration.
- Wipe the plate clean; every element should earn its place.

Common mistakes to avoid

- Overcrowding the plate so nothing stands out.
- Drowning the dish in sauce instead of placing it.
- Garnishing with inedible or pointless flourishes.

PRACTICE

Plate three dishes with a clear focal point, intentional sauce, and clean negative space. Photograph each and critique it.

Lesson 12.3 - The Brigade and Kitchen Roles

Objective: Understand how professional kitchens are organized.

The brigade de cuisine, formalized by Escoffier, is the hierarchy that lets a busy kitchen run smoothly. The executive chef (chef de cuisine) leads; the sous chef is second-in-command; chefs de partie run stations (saucier for sauces, poissonnier for fish, garde manger for cold dishes, patissier for pastry); commis and stagiaires assist and learn.

Even small kitchens borrow this logic: clear stations, clear responsibilities, and a culture of 'yes, chef' communication that keeps service coordinated under pressure. Understanding the structure helps you know where you would start, how you would advance, and how a kitchen actually functions as a team.

Key points

- The brigade is the kitchen's hierarchy of stations and roles.
- Chef de cuisine leads; sous chef is second; chefs de partie run stations.
- Stations include saucier, poissonnier, garde manger, and patissier.
- Clear roles and crisp communication keep service coordinated.

Common mistakes to avoid

- Assuming a kitchen is a free-for-all rather than a structured team.
- Not knowing the common station names when you walk in.
- Underestimating how much teamwork drives a service.

PRACTICE

Map a simple home version of the brigade to your capstone meal: assign yourself stations and a firing order as if you were a small team.

Lesson 12.4 - Your Career Path

Objective: Plan a realistic route into professional cooking.

There are many roads into the industry, and culinary school is only one of them. The most common entry is starting low (commis, prep, or a stage - an unpaid working trial) and learning on the line; many excellent chefs are self-taught and grew through experience, curiosity, and relentless practice. What matters is demonstrated skill and reliability.

Build a path you can actually walk: keep drilling fundamentals, document your cooking as a portfolio (photos, menus, recipes), seek out a stage at a kitchen you admire, and consider targeted certifications like food-safety credentials. Decide what 'professional chef' means to you - restaurant line, private chef, caterer, or your own kitchen - and aim there deliberately.

Key points

- Many chefs are self-taught; demonstrated skill and reliability matter most.
- A stage (working trial) is a common way in - and a way to learn.
- Build a portfolio: photos, menus, and recipes of your work.
- Pick a target - line cook, private chef, caterer - and plan toward it.

Common mistakes to avoid

- Waiting to feel 'ready' instead of seeking real kitchen experience.
- No portfolio or record of what you can do.
- Drifting without a concrete next step or target role.

PRACTICE

Draft a 12-month plan: skills to drill, a target stage or role, a portfolio outline, and any certifications you want to earn.